

**Physics 7501 Quiz 2**  
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**(1.2.1) Pauli matrices**

What are the eigenvalues of the  $2 \times 2$  matrix  $e^{i\theta\hat{Z}}$  where  $\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  is the 3rd Pauli matrix?

For a diagonal matrix  $M$ , we exponentiate every element along the main diagonal to get

$$e^M = \begin{pmatrix} e^{M_{11}} & & 0 \\ & \ddots & \\ 0 & & e^{M_{nn}} \end{pmatrix}$$

thus

$$e^{i\theta\hat{Z}} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

Since this is a diagonal matrix, its eigenvalues are just the diagonal entries  $e^{\pm i\theta}$

**(1.2.2) Linear algebra**

Find the simultaneous (i.e. common) eigenkets of Pauli matrices  $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ .  
If you cannot find any, explain.

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**(1.2.3) Measurements**

Consider the following normalized state of a single qubit:

$$|\psi\rangle = \begin{pmatrix} \cos \theta e^{i\phi} \\ \sin \theta \end{pmatrix}. \tag{1.3}$$

For a measurement of Pauli operator  $\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , what is the probability  $\mathbf{P}(X = +1)$  of measurement outcome  $X = 1$ ? Express it as a function of  $\theta$  and  $\phi$ .

In the standard  $\hat{Z}$ -basis, the  $X = +1$  state is given by

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The Born rule states that  $\mathbf{P}(X = +1) = |\langle + | \psi \rangle|^2$ , thus

$$\langle + | \psi \rangle = \frac{1}{\sqrt{2}} (\cos \theta e^{i\phi} + \sin \theta)$$

$\Rightarrow$

$$\begin{aligned} \mathbf{P}(X = +1) &= |\langle + | \psi \rangle|^2 = \frac{1}{2} (\cos^2 \phi + (e^{-i\phi} + e^{i\theta}) \sin \theta \cos \theta + \sin^2 \theta) \\ &= \frac{1}{2} (1 + 2 \sin \theta \cos \theta \cos \phi) \\ &= \frac{1}{2} (1 + \sin(2\theta) \cos \phi) \end{aligned}$$